

Business Problem:

The retail company has observed stagnant growth and declining customer engagement metrics over the past quarters. Initial assessments indicate potential issues in product performance variability, ineffective customer segmentation, and lack of insights into customer purchasing behavior. The company seeks to leverage its sales transaction data, customer profiles, and product inventory information to address the following key business problems:

- **Product Performance Variability:** Identifying which products are performing well in terms of sales and which are not. This insight is crucial for inventory management and marketing focus.
- **Customer Segmentation:** The company lacks a clear understanding of its customer base segmentation. Effective segmentation is essential for targeted marketing and enhancing customer satisfaction.
- **Customer Behaviour Analysis:** Understanding patterns in customer behavior, including repeat purchases and loyalty indicators, is critical for tailoring customer engagement strategies and improving retention rates.

Project Objectives

1. Analyze product performance to identify high-performing and low-performing products based on sales, profit, and customer demand.
2. Understand customer segmentation by categorizing customers into different value groups to support targeted marketing strategies.
3. Evaluate customer purchasing behavior by analyzing repeat purchases, order frequency, and average order value.
4. Identify revenue and profit trends across products, categories, and regions to support business growth decisions.
5. Improve customer retention insights by detecting loyalty patterns and repeat customer contributions.
6. Develop an interactive Power BI dashboard to provide actionable insights for management and support data-driven decision-making.
7. Generate business recommendations for improving inventory management, customer engagement, and overall sales performance.